Experiment 05 — Density-aware topic tree

arxivkg · preprocessing · phase 5 (topic tree, offline-browse artifact)

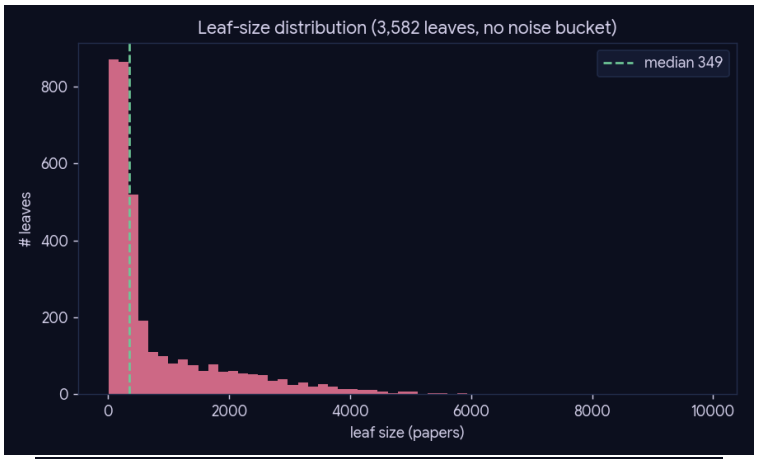
What we did

Show the shape of the shipped bisecting topic tree (`leaf` + `paper_topic`): no noise bucket (every paper gets a leaf), the leaf-size and depth distributions, and how coherence (leaf radius) tracks leaf size — i.e. the coherence early-stop keeps leaves tight while adapting grain to local density. Pure SQL + figures, no LLM; cached to `cache/summary.json`.

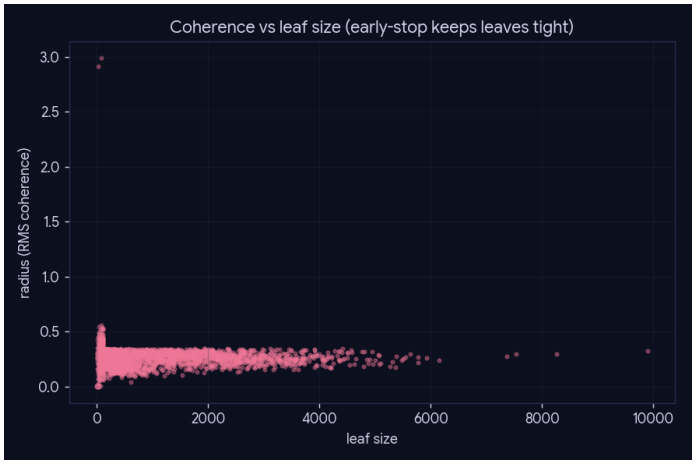
Summary

metric	value
leaves	3,582
papers assigned	2,918,448
leaf size min/median/p90/max	6 / 349 / 2348 / 9895

3,582 leaves over 2,918,448 papers (median leaf size 349, max 9895).







Conclusion

The tree places **all 2.92M** papers into **3,582** leaves (median size 349) with **no noise bucket** — the structural win over the DBSCAN-style design that dumped ∼355k niche papers into a "Miscellaneous" sink (experiment 04). Depth is non-uniform (deeper where dense, shallow where already coherent), and the coherence-vs-size scatter confirms the early-stop holds leaf radius down even as leaves grow, so a topic anchor lands a query in a tight, on-topic neighborhood. Leaf membership is from the pre-delta build (2025/2026 papers handled at serve time by `frontier_pull`); under the retrieval-only redesign the tree is retained only as an offline-browse artifact.